

Home Conceptual Laboratory Activity 9

At this time, you are requested to perform this activity at home in order to verify the concepts.

Objective: Verify how current-carrying conductor moves when passing through magnetic field.

Materials: Computer with internet connection.

Procedure:

watch this video and your observations.

<https://www.youtube.com/watch?v=F1PWnu01IQg>

Observations:

- A current-carrying copper tube in a magnetic field experiences a force.
- With one current direction, the tube is pushed inward toward the magnet.
- Reversing the current reverses the force, pushing the tube outward.
- The movement confirms the interaction between current and magnetic field.

Conclusion: A conductor carrying current in a magnetic field experiences a mechanical force. The direction of this force depends on the current's orientation and can be predicted using the right-hand rule. This principle explains how electric motors and magnetic levitation work.

Documentation:

